Matrix Decompositions

17 Fully-Solved Practice Problems

MFML / MFDS (AIMLCZC416 / DSECL ZC416)

Lecture 4 — Eigenvalues & Eigenvectors • Lecture 5 — Singular Value Decomposition

How to use this set. This set builds fluency in **matrix decompositions**: eigenvalues & eigenvectors, algebraic/geometric multiplicity, diagonalization, the eigendecomposition $A = PDP^{-1}$, the spectral (orthogonal) decomposition of symmetric matrices, the Singular Value Decomposition, low-rank approximation and the spectral norm. Read the two **concept boxes** below first — they explain the ideas the problems rely on. Then attempt each problem yourself before reading the **step-by-step** solution; every final result is in a green box.

Concept 1 • Algebraic vs. geometric multiplicity

Let λ be an eigenvalue of an $n \times n$ matrix A.

Algebraic multiplicity $\mathrm{am}(\lambda)$ — how many times λ occurs as a root of the characteristic polynomial $\det(A - \lambda I)$. (E.g. a factor $(\lambda - 3)^2$ means $\mathrm{am}(3) = 2$.)

Geometric multiplicity $\mathrm{gm}(\lambda)$ — the number of *independent* eigenvectors for λ, i.e. the dimension of the eigenspace:

$$\mathrm{gm}(\lambda) = \dim\ker(A - \lambda I) = n - \mathrm{rank}(A - \lambda I).$$

The key facts. (i) $1 \le \mathrm{gm}(\lambda) \le \mathrm{am}(\lambda)$ for every eigenvalue. (ii) A is **diagonalizable** $\iff$ $\mathrm{gm}(\lambda) = \mathrm{am}(\lambda)$ for *every* eigenvalue (enough eigenvectors to fill a basis). (iii) If an eigenvalue has $\mathrm{gm} < \mathrm{am}$, the matrix is *defective* and cannot be diagonalized.

One-line illustration. $\begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$ has $\lambda = 5$ with $\mathrm{am} = 2$ but $\mathrm{gm} = 2 - \mathrm{rank}\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) = 2 - 1 = 1$ — *defective*. By contrast $\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} = 5I$ has $\mathrm{am} = \mathrm{gm} = 2$ — *diagonalizable* (already diagonal).

Concept 2 • The eigenvalue toolkit (used in several problems)

If $A\boldsymbol{v} = \lambda\boldsymbol{v}$ (with $\boldsymbol{v} \neq \boldsymbol{0}$), then the eigenvalues transform predictably — *the eigenvector stays the same*:

Powers: $A^k\boldsymbol{v} = \lambda^k\boldsymbol{v} \quad \Rightarrow \quad \lambda^k$ is an eigenvalue of A^k.

Polynomials: if $p(t) = c_0 + c_1 t + \cdots + c_m t^m$, then $p(A)\boldsymbol{v} = p(\lambda)\boldsymbol{v}$.

Inverse: if A is invertible $(\lambda \neq 0)$, $A^{-1}\boldsymbol{v} = \frac{1}{\lambda}\boldsymbol{v}$.

Transpose: A^{T} has the same eigenvalues as A.

Two consequences let you avoid brute-force matrix multiplication. Writing the eigenvalues as $\lambda_1, \ldots, \lambda_n$:

$$\boxed{\mathrm{tr}(A) = \sum_i \lambda_i, \qquad \det(A) = \prod_i \lambda_i}$$

$$\Longrightarrow \qquad \mathrm{tr}(A^k) = \sum_i \lambda_i^k, \qquad \det(A^k) = \left(\det A\right)^k = \prod_i \lambda_i^k.$$

So to handle a high power, *first find the eigenvalues*, then work entirely with the scalars λ_i.

#	Problem theme	Core skill
1	Algebraic & geometric multiplicity (warm-up)	read off am, gm; decide diagonalizability
2	Eigen-analysis of a 3×3 (repeated λ)	char. polynomial; am vs. gm
3	Complex eigenvalues & eigenvectors	complex roots of a real 2×2
4	Parametric eigen-analysis	eigenpairs in symbols; when diagonalizable?
5	Is the matrix diagonalizable? (3×3)	multiplicity test; defective vs. not
6	Powers without brute force: det, tr	$\det(A^k) = (\det A)^k$, $\mathrm{tr}(A^k) = \sum \lambda_i^k$
7	Eigenvalues of a matrix polynomial $p(A)$	spectral mapping $\lambda \mapsto p(\lambda)$
8	Eigendecomposition $A = PDP^{-1}$	build P, D for a non-symmetric matrix; verify
9	Spectral decomposition $A = PDP^{\mathsf{T}}$ (3×3)	symmetric matrix, orthonormal eigenbasis (irrational λ)
10	Matrix powers via diagonalization	$A^n = PD^nP^{\mathsf{T}}$
11	Full SVD of a 3×2 matrix	$U\Sigma V^{\mathsf{T}}$; orthogonal completion of U
12	Full SVD of a 2×3 matrix	3×3 $A^{\mathsf{T}}A$ with a zero eigenvalue
13	SVD with a zero singular value	rank-deficient; SVD vs. diagonalizability
14	Best rank-1 approximation	$\sigma_1 u_1 v_1^{\mathsf{T}}$ from a computed SVD
15	SVD as a sum of rank-1 matrices	$A = \sum \sigma_i u_i v_i^{\mathsf{T}}$; truncation
16	Spectral norm of a matrix	$\|A\|_2 = \sigma_{\max}$
17	Capstone: EVD vs. SVD of one matrix	how the two decompositions relate

Problem 1 • Algebraic & geometric multiplicity (warm-up) *[3 marks]*

For each matrix, state the eigenvalue(s), their *algebraic* multiplicity am and *geometric* multiplicity gm, and decide whether the matrix is diagonalizable.

$$A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \qquad C = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix}.$$

Solution

Matrix A. Triangular, so the only eigenvalue is $\lambda = 3$ with am $= 2$. gm $= 2 - \mathrm{rank}(A - 3I) = 2 - \mathrm{rank}\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) = 2 - 1 = 1$. Since gm $= 1 < 2 =$ am, A is **not** diagonalizable.

Matrix B. $B = 3I$: eigenvalue $\lambda = 3$ with am $= 2$, and $A - 3I = 0$ so gm $= 2 - 0 = 2$. Here gm $=$ am, so B **is** diagonalizable (already diagonal).

Matrix C. Triangular, eigenvalue $\lambda = 3$ with am $= 3$. $C - 3I = \left(\begin{smallmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ has rank 2, so gm $= 3 - 2 = 1 < 3$. **Not** diagonalizable (a single Jordan block).

Take-away: same eigenvalue, same am — but the number of independent eigenvectors (gm) is what decides diagonalizability.

Answer. A: $\lambda = 3$, am $= 2$, gm $= 1$ — not diagonalizable. B: $\lambda = 3$, am $=$ gm $= 2$ — diagonalizable. C: $\lambda = 3$, am $= 3$, gm $= 1$ — not diagonalizable.

Problem 2 • Eigen-analysis of a 3×3 matrix *[6 marks]*

Consider

$$A = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix}.$$

(a) Find the characteristic polynomial and all eigenvalues with their algebraic multiplicities. **(b)** For every eigenvalue find a basis of its eigenspace and state the geometric multiplicity. **(c)** Hence decide, with reason, whether A is diagonalizable.

Solution

(a) Characteristic polynomial. Note $A = 3I + J$, where J is the all-ones matrix. Expanding $\det(A - \lambda I)$ (subtract row 1 from rows 2 and 3, then expand) gives

$$\det(A - \lambda I) = -(\lambda - 6)(\lambda - 3)^2.$$

So $\lambda = 6$ (algebraic multiplicity 1) and $\lambda = 3$ (algebraic multiplicity 2).

(b) Eigenspaces. *For $\lambda = 6$:* $(A - 6I)\boldsymbol{v} = \mathbf{0}$ reads $-2x + y + z = 0$, $x - 2y + z = 0$, $x + y - 2z = 0$, whose solution is $x = y = z$. Basis $\{(1, 1, 1)^\mathsf{T}\}$, *geometric* multiplicity 1.

For $\lambda = 3$: $(A - 3I)\boldsymbol{v} = \mathbf{0}$ collapses to the single equation $x + y + z = 0$ (all three rows become identical). With two free variables,

$$\boldsymbol{v} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \qquad \text{geometric multiplicity 2.}$$

(c) Diagonalizable? For each eigenvalue, geometric multiplicity = algebraic multiplicity (1=1 and 2=2), so the three eigenvectors form a basis of $\mathbb{R}^3$ and A *is* diagonalizable. (In fact, A is symmetric, so the spectral theorem already guarantees this.)

Answer. $\det(A - \lambda I) = -(\lambda - 6)(\lambda - 3)^2$. $\lambda = 6$: mult. 1, eigenvector $(1, 1, 1)^\mathsf{T}$; $\lambda = 3$: mult. 2, eigenspace $\{x + y + z = 0\}$, basis $(-1, 1, 0)^\mathsf{T}, (-1, 0, 1)^\mathsf{T}$. A is diagonalizable.

Problem 3 • Complex eigenvalues of a real matrix *[4 marks]*

Find all eigenvalues and corresponding eigenvectors of

$$A = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix},$$

and explain geometrically why no real eigenvector exists.

Solution

Step 1 — Characteristic polynomial.

$$\det(A - \lambda I) = \det \begin{pmatrix} 3 - \lambda & -2 \\ 4 & -1 - \lambda \end{pmatrix} = (3 - \lambda)(-1 - \lambda) + 8 = \lambda^2 - 2\lambda + 5 = 0.$$

$$\lambda = \frac{2 \pm \sqrt{4-20}}{2} = 1 \pm 2i.$$

Step 2 — Eigenvectors. For $\lambda = 1+2i$, the first row of $(A-\lambda I)\boldsymbol{v} = \mathbf{0}$ gives $(3-\lambda)x - 2y = 0$, i.e. $(2-2i)x - 2y = 0 \Rightarrow y = (1-i)x$. Taking $x = 1$,

$$\boldsymbol{v}_1 = \begin{pmatrix} 1 \\ 1-i \end{pmatrix}.$$

Its conjugate $\boldsymbol{v}_2 = (1,\ 1+i)^\mathsf{T}$ belongs to $\lambda = 1-2i$.

Step 3 — Geometric reason. The discriminant is negative, so the eigenvalues form a complex-conjugate pair. Such a real 2×2 matrix acts as a *rotation combined with scaling*; a rotation maps no nonzero real vector onto a multiple of itself, hence there is no real eigenvector.

Answer. $\lambda = 1 \pm 2i$ with eigenvectors $\boldsymbol{v} = (1,\ 1 \mp i)^\mathsf{T}$. No real eigenvector exists because the eigenvalues are non-real (the map rotates the plane).

Problem 4 • Parametric eigen-analysis *[4 marks]*

Let $a, b, c \in \mathbb{R}$ with $b \neq 0$ and consider the upper-triangular matrix

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix}.$$

(a) Find the eigenvalues and an eigenvector for each (in terms of a, b, c). **(b)** State, with justification, exactly when A is diagonalizable.

Solution

(a) Eigenvalues and eigenvectors. For a triangular matrix the eigenvalues are the diagonal entries: $\lambda_1 = a,\ \lambda_2 = c$.

$\lambda_1 = a$: $(A - aI) = \begin{pmatrix} 0 & b \\ 0 & c-a \end{pmatrix}$. The equations are $by = 0$ and $(c-a)y = 0$; since $b \neq 0$ we get $y = 0$, x free, so $\boldsymbol{v}_1 = (1, 0)^\mathsf{T}$.

$\lambda_2 = c$ *(assume $c \neq a$)*: $(A - cI) = \begin{pmatrix} a-c & b \\ 0 & 0 \end{pmatrix}$ gives $(a-c)x + by = 0 \Rightarrow x = -\dfrac{b}{a-c}y$. Taking $y = a-c$, $\boldsymbol{v}_2 = (-b,\ a-c)^\mathsf{T}$.

(b) When is A diagonalizable? *Case $a \neq c$:* two distinct eigenvalues $\Rightarrow$ two independent eigenvectors $\Rightarrow$ diagonalizable.

Case $a = c$: then $\lambda = a$ has algebraic multiplicity 2, but $(A - aI) = \begin{pmatrix} 0 & b \\ 0 & 0 \end{pmatrix}$ has rank 1 (because $b \neq 0$), so the eigenspace is only 1-dimensional. Geometric mult. $1 < 2 \Rightarrow$ *not* diagonalizable.

Answer. $\lambda_1 = a$ with $\boldsymbol{v}_1 = (1, 0)^\mathsf{T}$; $\lambda_2 = c$ with $\boldsymbol{v}_2 = (-b,\ a-c)^\mathsf{T}$. A is diagonalizable $\iff a \neq c$ (when $a = c$ and $b \neq 0$ it is defective).

Problem 5 • Is the 3×3 matrix diagonalizable? *[4 marks]*

Decide whether

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

is diagonalizable. Justify using eigenvalues, algebraic and geometric multiplicities.

Solution

Step 1 — Eigenvalues. A is upper-triangular, so the eigenvalues are the diagonal entries: $\lambda = 2$ (algebraic multiplicity 2) and $\lambda = 3$ (algebraic multiplicity 1).

Step 2 — Eigenspace of the repeated eigenvalue.

$$A - 2I = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This has rank 2, so its null space has dimension $3 - 2 = 1$: solving gives $y = 0,\ z = 0,\ x$ free, i.e. eigenspace $\operatorname{span}\{(1, 0, 0)^{\mathsf{T}}\}$. *Geometric* multiplicity 1.

Step 3 — Compare. For $\lambda = 2$ the geometric multiplicity (1) is *less than* the algebraic multiplicity (2). Therefore A cannot have a basis of eigenvectors.

Answer. A is **not diagonalizable**: for $\lambda = 2$, geometric mult. $1 <$ algebraic mult. 2 (only one independent eigenvector $(1, 0, 0)^{\mathsf{T}}$). The block $\left[\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right]$ is a defective Jordan block.

Problem 6 • Powers without brute force: determinant and trace *[4 marks]*

Consider

$$A = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 0 & 3 \\ 1 & -1 & 4 \end{pmatrix}, \qquad \text{whose characteristic polynomial is } (\lambda - 1)(\lambda - 2)(\lambda - 3).$$

Without forming any matrix power explicitly, compute **(a)** $\det(A^8)$, **(b)** $\operatorname{tr}(A^6)$, and **(c)** $\operatorname{tr}\big(A^6 - 7A^2\big)$.

Solution

Step 0 — Read off the eigenvalues. The characteristic polynomial factors as $(\lambda-1)(\lambda-2)(\lambda-3)$, so the eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 3.$$

(Quick sanity check: $\operatorname{tr} A = 2 + 0 + 4 = 6 = 1 + 2 + 3$ and $\det A = 1 \cdot 2 \cdot 3 = 6$.)

(a) Determinant of a power. Since A^k has eigenvalues λ_i^k, its determinant is the product, i.e. $\det(A^k) = (\det A)^k$:

$$\det(A^8) = (\det A)^8 = 6^8 = 1\,679\,616.$$

(b) Trace of a power. $\operatorname{tr}(A^6) = \sum_i \lambda_i^6 = 1^6 + 2^6 + 3^6 = 1 + 64 + 729 = 794.$

(c) Trace of a polynomial in A. Trace is linear, and A^6, A^2 have eigenvalues λ_i^6, λ_i^2, so

$$\operatorname{tr}\big(A^6 - 7A^2\big) = \sum_i \big(\lambda_i^6 - 7\lambda_i^2\big) = 794 - 7\,(1+4+9) = 794 - 98 = 696.$$

Every answer used only the three scalars λ_i — no 3×3 multiplication.

Answer. $\det(A^8) = 6^8 = 1\,679\,616$; $\operatorname{tr}(A^6) = 794$; $\operatorname{tr}(A^6 - 7A^2) = 696$.

Problem 7 • Eigenvalues of a matrix polynomial $p(A)$ *[4 marks]*

Let A be a 3×3 matrix whose eigenvalues are $1,\ 2,\ 4$, and define

$$B = A^2 - 5A + 6I.$$

(a) Show that if λ is an eigenvalue of A (with eigenvector $\boldsymbol{v}$), then $\lambda^2 - 5\lambda + 6$ is an eigenvalue of B (with the *same* eigenvector). **(b)** Hence find the eigenvalues of B, then $\det(B)$ and $\operatorname{tr}(B)$, and state whether B is invertible. **(c)** For which eigenvalues of A would B fail to be invertible?

Solution

(a) Spectral mapping. Suppose $A\boldsymbol{v} = \lambda\boldsymbol{v}$. Then $A^2\boldsymbol{v} = A(\lambda\boldsymbol{v}) = \lambda^2\boldsymbol{v}$, and so

$$B\boldsymbol{v} = (A^2 - 5A + 6I)\boldsymbol{v} = \lambda^2\boldsymbol{v} - 5\lambda\boldsymbol{v} + 6\boldsymbol{v} = (\lambda^2 - 5\lambda + 6)\boldsymbol{v}.$$

Thus $\boldsymbol{v}$ is still an eigenvector, now with eigenvalue $p(\lambda) = \lambda^2 - 5\lambda + 6$.

(b) Eigenvalues of B and consequences. Apply $p(\lambda) = \lambda^2 - 5\lambda + 6$ to each eigenvalue of A:

$$p(1) = 1 - 5 + 6 = 2, \qquad p(2) = 4 - 10 + 6 = 0, \qquad p(4) = 16 - 20 + 6 = 2.$$

So B has eigenvalues $\{2,\ 0,\ 2\}$. Therefore

$$\det(B) = 2 \cdot 0 \cdot 2 = 0, \qquad \operatorname{tr}(B) = 2 + 0 + 2 = 4.$$

Because 0 is an eigenvalue (equivalently $\det B = 0$), B is **not invertible** — even though A itself is (none of its eigenvalues is 0).

(c) When is B singular? B is singular exactly when $p(\lambda) = 0$ for some eigenvalue λ of A. Since $p(\lambda) = (\lambda - 2)(\lambda - 3)$, this happens iff A has 2 or 3 as an eigenvalue. Here $\lambda = 2$ occurs, which is why B is singular.

Remark (Cayley–Hamilton). If A's characteristic polynomial were *equal* to $p(\lambda) = \lambda^2 - 5\lambda + 6$ — e.g. for $A = \left(\begin{smallmatrix} 0 & 1 \\ -4 & 5 \end{smallmatrix}\right)$ with eigenvalues $1, 4$ and char. poly $\lambda^2 - 5\lambda + 4$ — then $p(A)$ collapses to a multiple of I: indeed $A^2 - 5A + 4I = 0$ gives $A^2 - 5A + 6I = 2I$.

Answer. $p(\lambda) = \lambda^2 - 5\lambda + 6$ maps each λ; eigenvalues of B are $2, 0, 2$. $\det(B) = 0$, $\operatorname{tr}(B) = 4$, so B is **not invertible**. B is singular iff A has eigenvalue 2 or 3 (the roots of p).

Problem 8 • Eigendecomposition $A = PDP^{-1}$ *[4 marks]*

Find an eigendecomposition $A = PDP^{-1}$ of the non-symmetric matrix

$$A = \begin{pmatrix} 5 & -1 \\ 2 & 2 \end{pmatrix},$$

giving $P,\ D,\ P^{-1}$ explicitly, and verify the factorization.

Solution

Step 1 — Eigenvalues.

$$\det(A-\lambda I)=(5-\lambda)(2-\lambda)+2=\lambda^2-7\lambda+12=(\lambda-3)(\lambda-4)=0,$$

so $\lambda_1=3,\ \lambda_2=4$.

Step 2 — Eigenvectors. $\lambda_1=3$: $(A-3I)=\begin{pmatrix}2&-1\\2&-1\end{pmatrix}\Rightarrow 2x=y$, so $\boldsymbol{p}_1=(1,2)^\mathsf{T}$.

$\lambda_2=4$: $(A-4I)=\begin{pmatrix}1&-1\\2&-2\end{pmatrix}\Rightarrow x=y$, so $\boldsymbol{p}_2=(1,1)^\mathsf{T}$.

Step 3 — Assemble and verify.

$$P=\begin{pmatrix}1&1\\2&1\end{pmatrix},\quad D=\begin{pmatrix}3&0\\0&4\end{pmatrix},\quad \det P=-1,\quad P^{-1}=\begin{pmatrix}-1&1\\2&-1\end{pmatrix}.$$

Check: $PD=\begin{pmatrix}3&4\\6&4\end{pmatrix}$, and $PDP^{-1}=\begin{pmatrix}3&4\\6&4\end{pmatrix}\begin{pmatrix}-1&1\\2&-1\end{pmatrix}=\begin{pmatrix}5&-1\\2&2\end{pmatrix}=A$. ✓

Answer. $A=PDP^{-1}$ with $P=\begin{pmatrix}1&1\\2&1\end{pmatrix}$, $D=\begin{pmatrix}3&0\\0&4\end{pmatrix}$, $P^{-1}=\begin{pmatrix}-1&1\\2&-1\end{pmatrix}$.

Problem 9 • Spectral decomposition $A=PDP^\mathsf{T}$ of a symmetric 3×3 *[6 marks]*

For the symmetric tridiagonal matrix

$$A=\begin{pmatrix}2&1&0\\1&2&1\\0&1&2\end{pmatrix},$$

find its eigenvalues, an *orthonormal* set of eigenvectors, and hence write $A=PDP^\mathsf{T}$ with P orthogonal.

Solution

Step 1 — Eigenvalues. Expanding $\det(A-\lambda I)$ along the first row,

$$\det(A-\lambda I)=(2-\lambda)\big[(2-\lambda)^2-1\big]-1\cdot(2-\lambda)=(2-\lambda)\big[(2-\lambda)^2-2\big]=0.$$

Hence $\lambda=2$, and $(2-\lambda)^2=2\Rightarrow\lambda=2\pm\sqrt{2}$. The eigenvalues are

$$\lambda_1=2+\sqrt{2},\qquad \lambda_2=2,\qquad \lambda_3=2-\sqrt{2}.$$

Step 2 — Eigenvectors. $\lambda=2$: $(A-2I)\boldsymbol{v}=\mathbf{0}$ gives $y=0$ and $x+z=0$, so $\boldsymbol{v}_2=(1,0,-1)^\mathsf{T}$.

$\lambda = 2+\sqrt{2}$: the first row $-\sqrt{2}\,x + y = 0 \Rightarrow y = \sqrt{2}\,x$, and the third row $y - \sqrt{2}\,z = 0 \Rightarrow x = z$. Thus $\boldsymbol{v}_1 = (1,\ \sqrt{2},\ 1)^\mathsf{T}$.

$\lambda = 2-\sqrt{2}$: similarly $y = -\sqrt{2}\,x$ and $x = z$, giving $\boldsymbol{v}_3 = (1,\ -\sqrt{2},\ 1)^\mathsf{T}$.

The three eigenvectors are mutually orthogonal (guaranteed by the spectral theorem).

Step 3 — Normalise and assemble. $\|\boldsymbol{v}_2\| = \sqrt{2}$ and $\|\boldsymbol{v}_1\| = \|\boldsymbol{v}_3\| = 2$, so

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{\sqrt{2}}{2} & 0 & -\frac{\sqrt{2}}{2} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{pmatrix}, \qquad D = \begin{pmatrix} 2+\sqrt{2} & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2-\sqrt{2} \end{pmatrix}, \qquad A = PDP^\mathsf{T}.$$

Since the columns are orthonormal, $P^\mathsf{T}P = I$, i.e. $P^{-1} = P^\mathsf{T}$.

Answer. Eigenvalues $2+\sqrt{2},\ 2,\ 2-\sqrt{2}$ with orthonormal eigenvectors $\frac{1}{2}(1,\sqrt{2},1)^\mathsf{T}$, $\frac{1}{\sqrt{2}}(1,0,-1)^\mathsf{T}$, $\frac{1}{2}(1,-\sqrt{2},1)^\mathsf{T}$; $A = PDP^\mathsf{T}$ with P orthogonal and $D = \operatorname{diag}(2+\sqrt{2},\ 2,\ 2-\sqrt{2})$.

Problem 10 • Matrix powers via diagonalization *[4 marks]*

Using diagonalization, derive a closed form for A^n and evaluate A^4, where

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

Solution

Step 1 — Diagonalize (orthogonally). $\det(A-\lambda I) = (2-\lambda)^2 - 1 = (\lambda-1)(\lambda-3) = 0 \Rightarrow \lambda_1 = 1,\ \lambda_2 = 3$. Orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$ and $\frac{1}{\sqrt{2}}(1,-1)^\mathsf{T}$ give

$$P = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}, \qquad A = PDP^\mathsf{T}.$$

Step 2 — Power formula. Since $A = PDP^\mathsf{T}$ with P orthogonal, $A^n = PD^nP^\mathsf{T}$ and only the diagonal is raised:

$$A^n = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 3^n \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1+3^n & 1-3^n \\ 1-3^n & 1+3^n \end{pmatrix}.$$

Step 3 — Evaluate $n = 4$. $3^4 = 81$, so $A^4 = \frac{1}{2}\begin{pmatrix} 82 & -80 \\ -80 & 82 \end{pmatrix} = \begin{pmatrix} 41 & -40 \\ -40 & 41 \end{pmatrix}$.

Answer. $A^n = \frac{1}{2}\begin{pmatrix} 1+3^n & 1-3^n \\ 1-3^n & 1+3^n \end{pmatrix}$, $\qquad A^4 = \begin{pmatrix} 41 & -40 \\ -40 & 41 \end{pmatrix}$.

Problem 11 • Full SVD of a 3×2 matrix *[6 marks]*

Find a full singular value decomposition $A = U\Sigma V^\mathsf{T}$ of

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix},$$

including the third column of U obtained by orthogonal completion.

Solution

Step 1 — $A^\mathsf{T} A$, singular values, right singular vectors.

$$A^\mathsf{T} A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Eigenvalues: $\det\left(\begin{smallmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{smallmatrix}\right) = (2-\lambda)^2 - 1 = 0 \Rightarrow \lambda = 3, 1$. Thus $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$, with

$$\boldsymbol{v}_1 = \tfrac{1}{\sqrt{2}}(1,1)^\mathsf{T} \ (\lambda = 3), \qquad \boldsymbol{v}_2 = \tfrac{1}{\sqrt{2}}(1,-1)^\mathsf{T} \ (\lambda = 1).$$

Step 2 — Left singular vectors $\boldsymbol{u}_i = \frac{1}{\sigma_i} A\boldsymbol{v}_i$.

$$\boldsymbol{u}_1 = \frac{1}{\sqrt{3}} A \, \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \qquad \boldsymbol{u}_2 = \frac{1}{1} A \, \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

Step 3 — Orthogonal completion $\boldsymbol{u}_3$. We need a unit vector with $\boldsymbol{u}_3 \perp \boldsymbol{u}_1$ and $\boldsymbol{u}_3 \perp \boldsymbol{u}_2$. Writing $\boldsymbol{u}_3 = (p,q,r)^\mathsf{T}$: from $\boldsymbol{u}_2$, $q - r = 0 \Rightarrow q = r$; from $\boldsymbol{u}_1$, $2p + q + r = 0 \Rightarrow p = -q$. So $\boldsymbol{u}_3 = \frac{1}{\sqrt{3}}(-1,1,1)^\mathsf{T}$.

Step 4 — Assemble.

$$U = \begin{pmatrix} 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad V^\mathsf{T} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Answer. $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$; $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$, $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^\mathsf{T}$; $\boldsymbol{u}_1 = \frac{1}{\sqrt{6}}(2,1,1)^\mathsf{T}$, $\boldsymbol{u}_2 = \frac{1}{\sqrt{2}}(0,1,-1)^\mathsf{T}$, $\boldsymbol{u}_3 = \frac{1}{\sqrt{3}}(-1,1,1)^\mathsf{T}$; $\Sigma = \left[\begin{smallmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$.

Problem 12 • Full SVD of a 2×3 matrix *[6 marks]*

Find the SVD of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

($Hint:$ it is quicker to get U and the singular values from the 2×2 matrix AA^T, then recover V from $\boldsymbol{v}_i = \frac{1}{\sigma_i} A^\mathsf{T} \boldsymbol{u}_i$; the missing right vector spans $\ker(A^\mathsf{T} A)$.)

Solution

Step 1 — AA^T, singular values, left singular vectors.

$$AA^\mathsf{T} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \Rightarrow \lambda = 3, 1, \quad \sigma_1 = \sqrt{3},\ \sigma_2 = 1.$$

$$\boldsymbol{u}_1 = \tfrac{1}{\sqrt{2}}(1,1)^\mathsf{T}\ (\lambda = 3), \qquad \boldsymbol{u}_2 = \tfrac{1}{\sqrt{2}}(1,-1)^\mathsf{T}\ (\lambda = 1).$$

Step 2 — Right singular vectors $\boldsymbol{v}_i = \frac{1}{\sigma_i} A^\mathsf{T} \boldsymbol{u}_i$.

$$\boldsymbol{v}_1 = \frac{1}{\sqrt{3}} A^\mathsf{T} \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad \boldsymbol{v}_2 = \frac{1}{1} A^\mathsf{T} \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Step 3 — Third right vector from the null space. A is 2×3, so Σ needs a third right vector $\boldsymbol{v}_3$ with singular value 0, spanning $\ker(A)$ (equivalently $\ker A^\mathsf{T} A$). Solving $A\boldsymbol{v} = \mathbf{0}$: $x + y = 0,\ y + z = 0$ gives $\boldsymbol{v}_3 = \frac{1}{\sqrt{3}}(1,-1,1)^\mathsf{T}$.

Step 4 — Assemble.

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix}.$$

The single zero column of Σ reflects that A has rank 2 but 3 columns.

Answer. $\sigma_1 = \sqrt{3}$, $\sigma_2 = 1$; $U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, $\Sigma = \left[\begin{smallmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right]$, $\boldsymbol{v}_1 = \frac{1}{\sqrt{6}}(1,2,1)^\mathsf{T}$, $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,0,-1)^\mathsf{T}$, $\boldsymbol{v}_3 = \frac{1}{\sqrt{3}}(1,-1,1)^\mathsf{T}$.

Problem 13 • SVD with a zero singular value; SVD vs. diagonalizability *[6 marks]*

(a) Find the SVD of $A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$. **(b)** Show that A is *not* diagonalizable, yet its SVD still exists, and explain why.

Solution

(a) Step 1 — Singular values.

$$A^\mathsf{T} A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}, \qquad \lambda = 4, 0 \Rightarrow \sigma_1 = 2,\ \sigma_2 = 0.$$

The single nonzero singular value shows rank $A = 1$.

Step 2 — Right and left vectors. Eigenvectors of $A^\mathsf{T}A$: $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$ (for 4), $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^\mathsf{T}$ (for 0).

$$\boldsymbol{u}_1 = \frac{1}{\sigma_1}A\boldsymbol{v}_1 = \frac{1}{2}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}2\\-2\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}, \qquad \boldsymbol{u}_2 \perp \boldsymbol{u}_1 : \ \boldsymbol{u}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}.$$

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\-1 & 1\end{pmatrix}, \quad \Sigma = \begin{pmatrix}2 & 0\\0 & 0\end{pmatrix}, \quad V = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}.$$

(b) Not diagonalizable, but SVD exists. The eigenvalues of A itself satisfy $\det(A-\lambda I) = \lambda^2 = 0$, so $\lambda = 0$ has algebraic multiplicity 2, while $A\boldsymbol{v} = \mathbf{0}$ has the single solution direction $(1,-1)^\mathsf{T}$ — geometric multiplicity 1. Since $1 < 2$, A has no eigenbasis and is **not diagonalizable**. Nevertheless the SVD is built from the *symmetric* matrices $A^\mathsf{T}A$ and AA^T, which are *always* orthogonally diagonalizable. This is exactly why an SVD exists for *every* matrix, diagonalizable or not.

Answer. $\sigma_1 = 2,\ \sigma_2 = 0;\ U = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\-1 & 1\end{pmatrix},\ \Sigma = \begin{pmatrix}2 & 0\\0 & 0\end{pmatrix},\ V = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\1 & -1\end{pmatrix}$. A is not diagonalizable ($\lambda = 0$: algebraic mult. 2, geometric mult. 1).

Problem 14 • Best rank-1 approximation *[4 marks]*

For $A = \begin{pmatrix}2 & 2\\-1 & 1\end{pmatrix}$, find the singular values and the best rank-1 approximation $A_1 = \sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^\mathsf{T}$.

Solution

Step 1 — Singular values and top right vector.

$$A^\mathsf{T}A = \begin{pmatrix}2 & -1\\2 & 1\end{pmatrix}\begin{pmatrix}2 & 2\\-1 & 1\end{pmatrix} = \begin{pmatrix}5 & 3\\3 & 5\end{pmatrix}, \qquad \det\begin{pmatrix}5-\lambda & 3\\3 & 5-\lambda\end{pmatrix} = (\lambda-8)(\lambda-2) = 0.$$

So $\sigma_1 = \sqrt{8} = 2\sqrt{2},\ \sigma_2 = \sqrt{2}$, with $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$ for $\lambda = 8$.

Step 2 — Top left vector.

$$\boldsymbol{u}_1 = \frac{1}{\sigma_1}A\boldsymbol{v}_1 = \frac{1}{2\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}4\\0\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}.$$

Step 3 — Rank-1 approximation.

$$A_1 = \sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^\mathsf{T} = 2\sqrt{2}\begin{pmatrix}1\\0\end{pmatrix}\cdot\tfrac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\end{pmatrix} = 2\begin{pmatrix}1 & 1\\0 & 0\end{pmatrix} = \begin{pmatrix}2 & 2\\0 & 0\end{pmatrix}.$$

By the Eckart–Young theorem this is the closest rank-1 matrix to A; the discarded part corresponds to the dropped singular value $\sigma_2 = \sqrt{2}$.

Answer. $\sigma_1 = 2\sqrt{2},\ \sigma_2 = \sqrt{2};\quad A_1 = \sigma_1\boldsymbol{u}_1\boldsymbol{v}_1^\mathsf{T} = \begin{pmatrix}2 & 2\\0 & 0\end{pmatrix}$ with $\boldsymbol{u}_1 = (1,0)^\mathsf{T},\ \boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$.

Problem 15 • SVD as a sum of rank-1 matrices *[4 marks]*

A matrix has the SVD data

$$\sigma_1 = 3,\ \boldsymbol{u}_1 = \tfrac{1}{\sqrt{2}}(1,1)^\mathsf{T},\ \boldsymbol{v}_1 = \tfrac{1}{\sqrt{2}}(1,1)^\mathsf{T}; \qquad \sigma_2 = 1,\ \boldsymbol{u}_2 = \tfrac{1}{\sqrt{2}}(1,-1)^\mathsf{T},\ \boldsymbol{v}_2 = \tfrac{1}{\sqrt{2}}(1,-1)^\mathsf{T}.$$

(a) Reconstruct $A = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^\mathsf{T} + \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^\mathsf{T}$. **(b)** Write down the best rank-1 approximation and the relative weight it keeps.

Solution

(a) Two rank-1 layers.

$$\sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^\mathsf{T} = 3 \cdot \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \sigma_2 \boldsymbol{u}_2 \boldsymbol{v}_2^\mathsf{T} = 1 \cdot \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

Adding,

$$A = \frac{1}{2} \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

(b) Best rank-1 approximation. Keep the largest singular value only:

$$A_1 = \sigma_1 \boldsymbol{u}_1 \boldsymbol{v}_1^\mathsf{T} = \frac{3}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

This is the rank-k truncation $\widehat{A}(k) = \sum_{i=1}^{k} \sigma_i \boldsymbol{u}_i \boldsymbol{v}_i^\mathsf{T}$ with $k = 1$. The energy kept is $\dfrac{\sigma_1^2}{\sigma_1^2 + \sigma_2^2} = \dfrac{9}{10} =$ 90%.

Answer. $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$; best rank-1 approximation $A_1 = \frac{3}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, keeping $\sigma_1^2/(\sigma_1^2 + \sigma_2^2) =$ 90% of the energy.

Problem 16 • Spectral norm of a matrix *[4 marks]*

Compute the spectral norm $\|A\|_2$ of

$$A = \begin{pmatrix} 4 & 3 \\ 0 & 5 \end{pmatrix},$$

and state the corresponding right singular vector.

Solution

Step 1 — Use $\|A\|_2 = \sigma_{\max} = \sqrt{\lambda_{\max}(A^\mathsf{T} A)}$.

$$A^\mathsf{T} A = \begin{pmatrix} 4 & 0 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 4 & 3 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 16 & 12 \\ 12 & 34 \end{pmatrix}.$$

Step 2 — Largest eigenvalue of $A^\mathsf{T} A$. $\det(A^\mathsf{T} A - \lambda I) = \lambda^2 - 50\lambda + 400 = 0$ (trace 50, determinant $16 \cdot 34 - 12^2 = 400$), so

$$\lambda = \frac{50 \pm \sqrt{2500 - 1600}}{2} = \frac{50 \pm 30}{2} = 40 \text{ or } 10.$$

Hence $\sigma_{\max} = \sqrt{40} = 2\sqrt{10}$ and $\sigma_{\min} = \sqrt{10}$.

Step 3 — Right singular vector for $\sigma_{\max}$. $(A^\mathsf{T}A - 40I)\boldsymbol{v} = \begin{pmatrix} -24 & 12 \\ 12 & -6 \end{pmatrix}\boldsymbol{v} = \mathbf{0} \Rightarrow 2x = y$, so $\boldsymbol{v}_1 = \frac{1}{\sqrt{5}}(1,2)^\mathsf{T}$.

$$\|A\|_2 = 2\sqrt{10} \approx 6.32.$$

Answer. $\|A\|_2 = \sigma_{\max} = 2\sqrt{10} \approx 6.32$, attained along $\boldsymbol{v}_1 = \frac{1}{\sqrt{5}}(1,2)^\mathsf{T}$ (the other singular value is $\sqrt{10}$).

Problem 17 • Capstone — eigendecomposition vs. SVD of one matrix *[6 marks]*

For $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$: **(a)** find its eigendecomposition (note the *negative* eigenvalue); **(b)** find its SVD; **(c)** explain precisely how the two are related.

Solution

(a) Eigendecomposition. $\det(A - \lambda I) = (1-\lambda)^2 - 4 = (\lambda - 3)(\lambda + 1) = 0 \Rightarrow \lambda_1 = 3,\ \lambda_2 = -1$. Orthonormal eigenvectors $\boldsymbol{q}_1 = \frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$ (for 3) and $\boldsymbol{q}_2 = \frac{1}{\sqrt{2}}(1,-1)^\mathsf{T}$ (for -1). With $Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$,

$$A = Q\begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}Q^\mathsf{T}.$$

(b) SVD. $A^\mathsf{T}A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ has eigenvalues 9 and 1, so $\sigma_1 = 3,\ \sigma_2 = 1$ — note $\sigma_i = |\lambda_i|$. The right singular vectors are the same eigenvectors $\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1,1)^\mathsf{T}$, $\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}(1,-1)^\mathsf{T}$, and

$$\boldsymbol{u}_1 = \frac{1}{3}A\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \boldsymbol{v}_1, \qquad \boldsymbol{u}_2 = \frac{1}{1}A\boldsymbol{v}_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = -\boldsymbol{v}_2.$$

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}, \quad V = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

(c) The relationship. For a symmetric matrix the singular values are the *absolute values* of the eigenvalues, $\sigma_i = |\lambda_i|$, and the right singular vectors are the eigenvectors. Where an eigenvalue is positive, $\boldsymbol{u}_i = \boldsymbol{v}_i$; where it is *negative* (here $\lambda_2 = -1$), the minus sign is absorbed into the left vector, $\boldsymbol{u}_i = -\boldsymbol{v}_i$. Thus EVD and SVD coincide exactly when every eigenvalue is non-negative (a positive-semidefinite matrix); otherwise they differ only by these sign flips between U and V.

Answer. EVD: $A = Q\,\mathrm{diag}(3,-1)Q^\mathsf{T}$, $Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. SVD: $\Sigma = \mathrm{diag}(3,1) = \mathrm{diag}(|\lambda_i|)$, $V = Q$, $U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$ (the negative eigenvalue flips the sign of $\boldsymbol{u}_2$ relative to $\boldsymbol{v}_2$).

End of practice set • 17 problems • all topics from Lecture 4 & Lecture 5 (decompositions)